

Rethash Reddy

Software Engineer (Distributed Systems, Security & Decentralized Platforms)

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Summary

Software engineer with hands-on experience building real-time and distributed systems, including AI-assisted tooling. Particularly interested in decentralized identity, secure communication, and trust-first infrastructure. Comfortable working across the stack, from low-level Unix processes to full-stack TypeScript applications, with a focus on reliability, security, and clean system design.

Skills

Core concepts: Distributed systems, real-time data pipelines, secure APIs, Unix system programming

Languages: TypeScript, JavaScript, C/C++, Python, Java

Web: React, Next.js, Node.js, Express, WebSockets

Databases: PostgreSQL, MongoDB, MySQL

Tools: Drizzle ORM, Git, Machine Learning (Statistical anomaly detection and clustering for telemetry data)

Strengths: Problem solving, competitive programming, leadership

Projects

Observify — Full-Stack Web Observability Platform

Built Observify, a real-time distributed observability platform for collecting telemetry from multiple clients. Implemented a TypeScript SDK and Node.js ingestion backend to capture errors, performance metrics, and network failures over persistent connections. Applied ML-based anomaly detection and built a React dashboard to visualize system health in real time.

Tech: TypeScript, Node.js, Express, React, Python

Clip-Based AI Video Generator

Built a clip-based video generation pipeline that transforms a single prompt into coherent long-form videos by orchestrating multiple short-generation runs. Designed deterministic prompt sequencing and automated clip assembly to maintain continuity under strict compute constraints. Implemented reliable stitching and post-processing using FFmpeg with a focus on reproducibility and efficiency.

Tech: Python, HuggingFace, FFmpeg

DecentraTrust — Decentralized Identity & Trust Infrastructure

Built a decentralized identity and trust system that lets organizations issue verifiable credentials which can be checked offline without contacting the issuer. Implemented a user-controlled wallet and a minimal frontend for viewing and presenting credentials, with trust rules and revocation enforced by policy; full cryptographic verification is handled in the core system, with frontend assumptions clearly documented as an MVP tradeoff.

Tech: ED25519, NodeJS and ES modules

AI-Enhanced Unix Shell

Built a custom Unix-style shell supporting command parsing, job control, and process execution using low-level system calls. Implemented foreground/background process management and signal handling to ensure reliable execution. Added lightweight AI-based typo correction using fuzzy matching to improve usability without compromising core shell behavior.

Tech: C, Unix System Programming

Education

B.E. in Computer Science Engineering
RV Institute of Technology and Management
CGPA: 8.5

Extracurricular

Organizing Committee — Robotics Club
Competitive Programming Club (Alcoholics)
CS Project Club